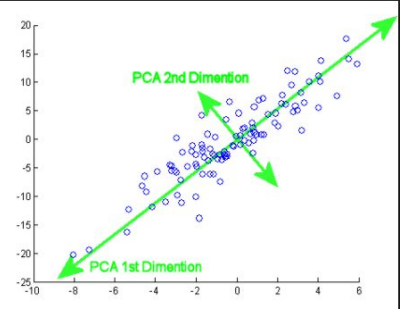




Principal Component Analysis and Singular Value Decomposition

Dimensionality reduction /
feature extraction

Principal Component Analysis

- Originated from the work by Pearson(1901).
- Its purpose is to derive new features (variables) in the decreasing order of importance.
- Dimensionality can be reduced without losing much information and structure present in the data.

PRINCIPAL COMPONENT ANALYSIS [PCA]

- * A method to reduce the dimensionality of the data.
- * PCA seeks a projection that best represents the data in a least squares sense.
- * In the new space data is so represented that the features become uncorrelated.
- * In the new space distributions might become a simpler one.

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- * In the new space distributions might become a simpler one.

Shortcomings

- * Needs to find covariance matrix and its eigen vectors
- * Discriminating components between classes might be lost

objective :-

Let $\mathcal{D} = \{x_1, \dots, x_n\}$ be the set of patterns of dimensionality d

We want to find $\mathcal{D}' = \{x'_1, x'_2, \dots, x'_n\}$

where each x'_i is of dim. d'

such that $d' < d$, and

$$J = \sum_{i=1}^n \|x_i - x'_i\|^2 \text{ should be minimum possible one.}$$

What is zero dim. projection for the data?

i.e., we want to represent the data set by just one pattern (x_0).

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$$\text{i.e., } x_0 = \frac{1}{n} \sum_i x_i$$

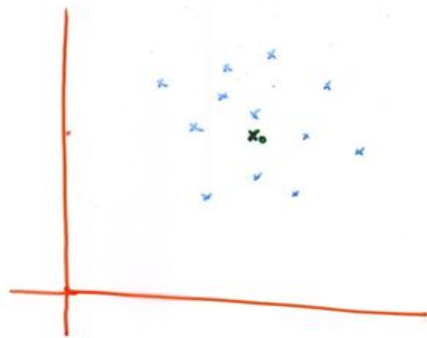
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What is good 1-Dim Representation

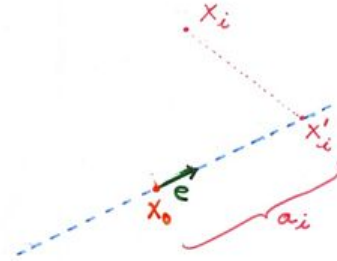
Let the data is projected onto a line
passing through the centroid (X_0)

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Let the data is projected onto a line
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$$\text{Let } X'_i = \text{Proj}(X_i)$$

$\vec{e}$ is unit vector
in the direction of
the line.



Then

$$X'_i = X_0 + a_i \vec{e}$$

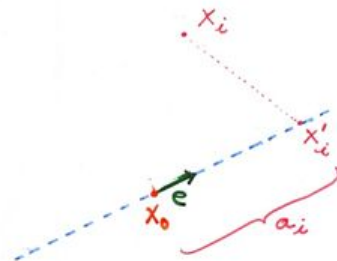
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$\rightarrow$ scalar
dist. between x_0 & x'_i

$$J = \sum_i \|(x_0 + a_i \vec{e}) - x_i\|^2$$

We need to find a_i & $\vec{e}$

To find $a_1, a_2, \dots, a_n$

$$\begin{aligned} J &= \sum_i \| (x_0 + a_i e) - x_i \|^2 \\ &= \sum_i \| a_i e - (x_i - x_0) \|^2 \\ &= \sum_i a_i^T e e^T - 2 \sum_i a_i^T e^T (x_i - x_0) + \sum_i \| x_i - x_0 \|^2 \\ &= \sum_i a_i^T - 2 \sum_i a_i^T e^T (x_i - x_0) + \sum_i \| x_i - x_0 \|^2 \end{aligned}$$

Now, To find a_j

$$\begin{aligned} \frac{\partial J}{\partial a_j} &= 2 a_j - 2 e^T (x_j - x_0) = 0 \\ a_j &= e^T (x_j - x_0) \end{aligned}$$

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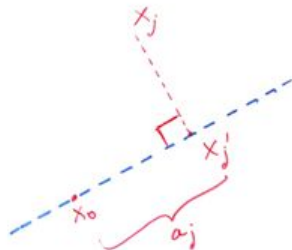
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$$\frac{\partial J}{\partial a_j} = 2 a_j - 2 e^T (x_j - x_0) = 0$$

$$a_j = e^T (x_j - x_0)$$

i.e., Perpendicularly
Project onto
the line



To Find $\vec{e}$

$$J = \sum a_i^2 - 2 \sum a_i e^T (x_i - x_0) + \sum \|x_i - x_0\|^2$$

$$\therefore a_i = e^T (x_i - x_0),$$

$$J = \sum a_i^2 - 2 \sum a_i^2 + \sum \|x_i - x_0\|^2$$

$$= - \sum a_i^2 + \sum \|x_i - x_0\|^2$$

$$= - \sum e^T (x_i - x_0) (x_i - x_0)^T e + \sum \|x_i - x_0\|^2$$

$$= - \left[e^T \left(\sum (x_i - x_0) (x_i - x_0)^T \right) e \right] + \sum \|x_i - x_0\|^2$$

$$= - e^T S e + \sum \|x_i - x_0\|^2$$

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$$= - e^T S e + \sum \|x_i - x_0\|^2$$

where $S = \text{Scatter matrix} = \sum_i (x_i - x_0) (x_i - x_0)^T$

$S = n \cdot (\text{sample Covariance Matrix})$

$$J = -e^T S e + \sum \|x_i - x_0\|^2$$

To minimize J , $-e^T S e$ should be minimized subject to the constraint

$$\|e\| = 1 \quad \text{or} \quad e^T e - 1 = 0$$

This is "Constrained Optimization" problem.

We can use the method of "Lagrange multipliers"

Constrained OptimizationMinimize $f(v)$ subject to $g_j(v) \leq 0$, for $1 \leq j \leq n$.

$$\text{Lagrangian, } \mathcal{L} = f(v) + \sum_{j=1}^n \alpha_j g_j(v)$$

$\downarrow$
 Lagrange Multiplier

Necessary cond. at optimal v are :

- (i) $\nabla_v \mathcal{L} = 0$
 - (ii) $\alpha_j \geq 0$
 - (iii) $\alpha_j g_j(v) = 0$
- } for all $j = 1$ to n

But, we are with equality constraint.

- So, gradient w.r.t. primal variables and gradient w.r.t. dual variables can be equated to zero.

Minimize $-e^T s e$
such that $e^T e - 1 = 0$

$$\mathcal{L} = (-e^T s e) + \alpha (e^T e - 1)$$

$$\nabla_e \mathcal{L} = \frac{\partial \mathcal{L}}{\partial e} = -2s e + 2\alpha e = 0$$

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$$s e = \alpha e$$

Diagram illustrating the equation $s e = \alpha e$ with annotations:

- s is annotated as "scatter matrix" (with a squiggly arrow pointing to s).
- e is annotated as "vector" (with a squiggly arrow pointing to e).
- α is annotated as "scalar" (with a squiggly arrow pointing to α).

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$$\mathcal{L} = (-e^T S e) + \alpha (e^T e - 1)$$

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$$S e = \alpha e$$

Scatter matrix $\downarrow$ scalar $\rightarrow$ vector

This is eigen value (vector) problem

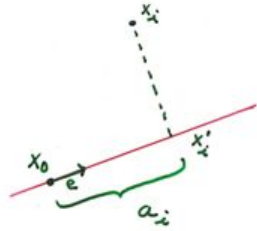
α is eigen value
 e is eigen vector } for S

$\therefore -e^T S e$ minimized $\Rightarrow -e^T \alpha e$ minimized
 $\Rightarrow \alpha$ should be maximum.

i.e., e is the eigen vector for S for which
eigen value is maximum.

e gives maximum Variance Direction

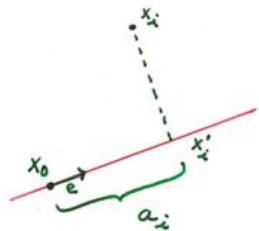
$$a_i = e^T (x_i - x_0)$$



$\{x_1, x_2, \dots, x_n\}$ is represented
as $\{a_1, a_2, \dots, a_n\}$

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Variance of $\{a_1, \dots, a_n\}$

$$= \frac{1}{n} \sum (a_i - a_0)^2 = \frac{1}{n} \sum a_i^2$$

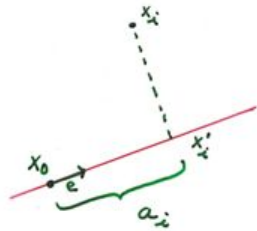
$[\because a_0 = 0]$

$$= \frac{1}{n} e^T \left[\sum (x_i - x_0)(x_i - x_0)^T \right] e$$

$$= \frac{1}{n} e^T S e$$

e gives maximum Variance Direction

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* The new space is so found that $e^T S e$ is maximum possible one.

i.e., The data is projected onto that line over which the variance is large.

Generalization : To find d' dimensions

$$x'_i = x_0 + [a_{i1} e_1 + \dots + a_{id'} e_{d'}]$$

We get, $J = \sum \left\| x_0 + \sum_{j=1}^{d'} a_{ij} e_j - x_i \right\|^2$

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It is easy to show that

$\left. \begin{matrix} e_1 \\ e_2 \\ \vdots \\ e_{d'} \end{matrix} \right\}$ are eigen vectors of S
for which eigen values are maximum possible.

$$S e_1 = \alpha_1 e_1$$

$$S e_2 = \alpha_2 e_2$$

$$\vdots$$

$$S e_{d'} = \alpha_{d'} e_{d'}$$

$$\alpha_1 \geq \alpha_2 \geq \dots \geq \alpha_{d'}$$

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Because the scatter matrix S is real and symmetric, Eigen values are real and nonnegative.

Representation in New Space

$$X'_i = X_0 + [a_{i1} e_1 + \dots + a_{id'} e_{d'}]$$

$\therefore X_0, e_1, \dots, e_{d'}$ are fixed, So

$$X'_i \text{ can be represented as } \begin{bmatrix} a_{i1} \\ a_{i2} \\ \vdots \\ a_{id'} \end{bmatrix} = Y_i$$

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$$\begin{aligned} \text{Since } a_{i1} &= e_1^T (X_i - X_0) \\ a_{i2} &= e_2^T (X_i - X_0) \\ &\vdots \\ a_{id'} &= e_{d'}^T (X_i - X_0) \end{aligned}$$

$$Y_i = \begin{bmatrix} e_1^T \\ e_2^T \\ \vdots \\ e_{d'}^T \end{bmatrix} (X_i - X_0)$$

 Transformation matrix

$$\text{Let us call this, } P = \begin{bmatrix} - e_1 - \\ - e_2 - \\ \vdots \\ - e_{d'} - \end{bmatrix}_{d' \times d}$$

The projection matrix P

- $P P' = I$, but P need not be square.
If P is square, i.e., use all eigenvectors, then P is orthogonal.
Not only P is orthogonal, P is a rotation matrix.

The projection matrix P

- $P P^T = I$, but P need not be square.

If P is square, i.e., use all eigenvectors, then P is orthogonal.

Not only P is orthogonal, P is a rotation matrix.

- PCA basically, translates (so that origin becomes centroid) and does rotation (*flip* is also included; so that features are uncorrelated).

Data

x	y
2.5	2.4
0.5	0.7
2.2	2.9
1.9	2.2
3.1	3.0
2.3	2.7
2	1.6
1	1.1
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Mean subtracted Data	
x	y
.69	.49
-1.31	-1.21
.39	.99
.09	.29
1.29	1.09
.49	.79
.19	-.31
-.81	-.81
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Eigenvalues are 1.28, 0.049

Corresponding Eigenvectors are $\begin{pmatrix} -.67 \\ -.73 \end{pmatrix}$ and $\begin{pmatrix} -.73 \\ .67 \end{pmatrix}$

What is the projection matrix?

If we are interested in 1D data

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What is the projection matrix?

$$P = \begin{pmatrix} -.67 \\ -.73 \end{pmatrix}^t$$

Transformed Data (Single eigenvector)

x
-.827970186
1.77758033
-.992197494
-.274210416
-1.67580142
-.912949103
.0991094375
1.14457216
.438046137
1.22382056

The data after transforming using only the most significant eigenvector

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2.2	2.9
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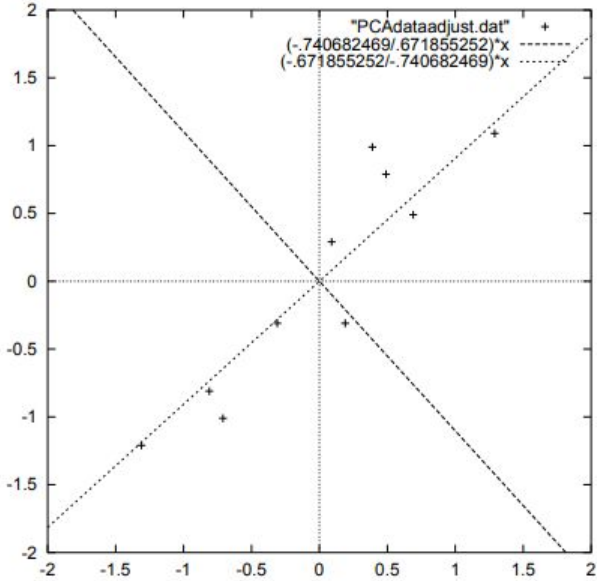
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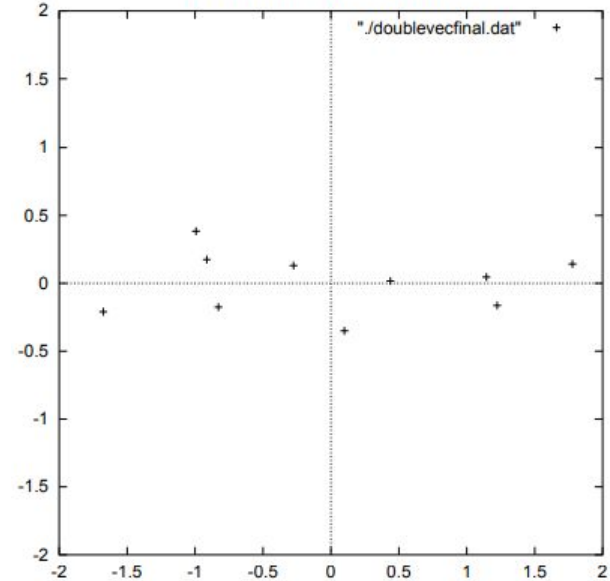
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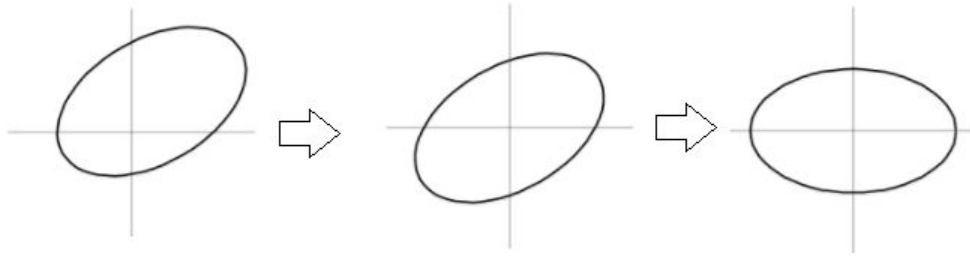
We can get a 2D data by using both eigenvectors,

Mean adjusted data with eigenvectors overlayed



Data transformed with 2 eigenvectors





**HOW IS THAT FEATURES ARE
UNCORRELATED IN THE NEW
SPACE?**

In the new space features are un-correlated!

$$Y_i = P(X_i - X_0) = PX_i - PX_0$$

$$\text{Mean in the new space, } Y_0 = P(X_0 - X_0) = \vec{0}$$

$$\text{Scatter matrix in new space} = S'$$

$$S' = \sum_i (Y_i - Y_0)(Y_i - Y_0)^T = \sum_i Y_i Y_i^T$$

$$= \sum_i (PX_i - PX_0)(PX_i - PX_0)^T$$

$$= \sum_i P(X_i - X_0)(X_i - X_0)^T P^T$$

$$= P S P^T \quad \text{S is the scatter matrix of original space}$$

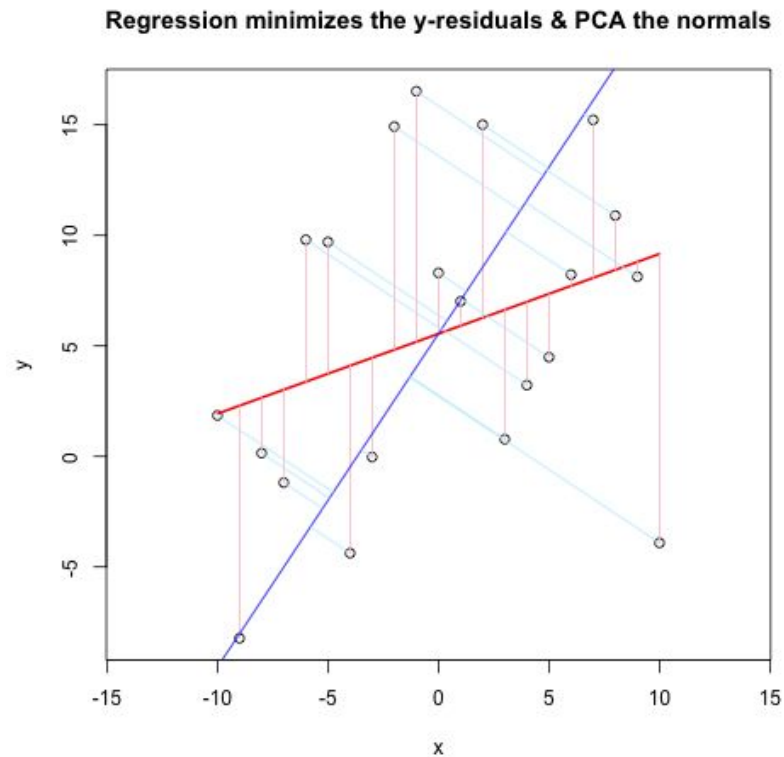
$$= \begin{bmatrix} -e_1 & - \\ -e_2 & - \\ \vdots & \\ -e_{d'} & - \end{bmatrix} S \begin{bmatrix} | & | & & | \\ e_1 & e_2 & \dots & e_{d'} \\ | & | & & | \end{bmatrix}$$

$$= \begin{bmatrix} -e_1 & - \\ -e_2 & - \\ \vdots & \\ -e_{d'} & - \end{bmatrix} [\alpha_1 e_1 \dots \alpha_{d'} e_{d'}]$$

$$= \begin{bmatrix} \alpha_1 & & & \\ & \alpha_2 & & \\ & & \ddots & \\ 0 & & & \alpha_{d'} \end{bmatrix}$$

- **Linear regression** also fitted a line, and, now, your **PCA** is also fitting a line (mostly), what are the differences??

Red: Linear
regression
Blue: PCA

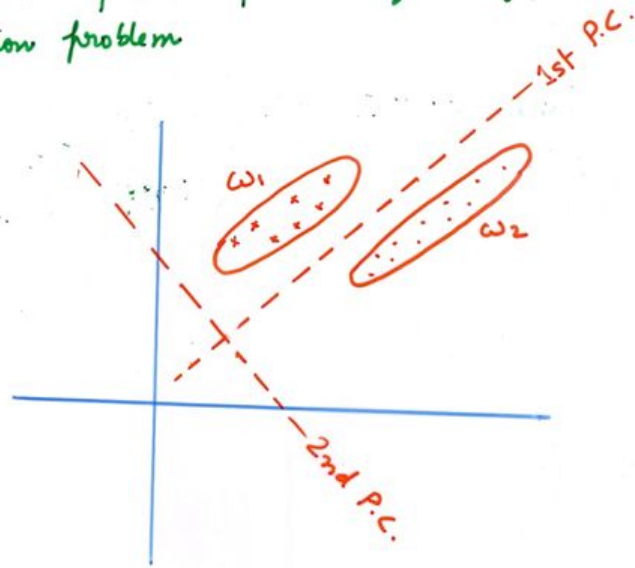


Drawback of PCA

* PCA seeks directions that are efficient for representation

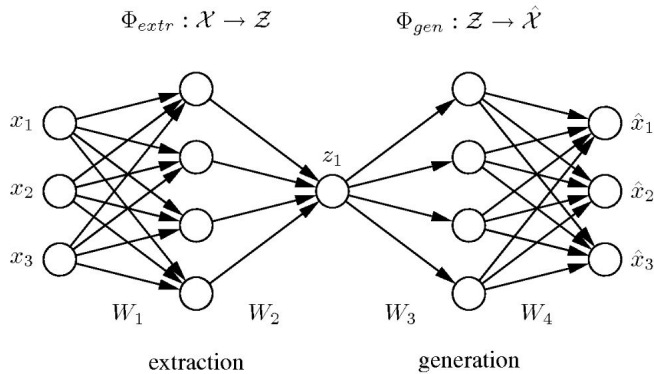
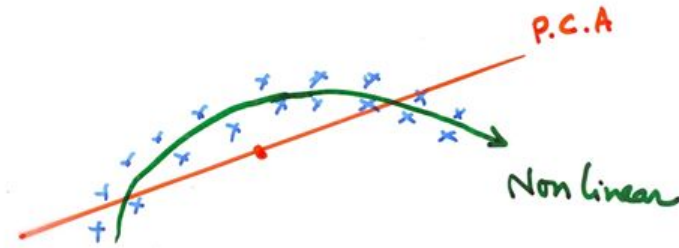
But does not take class-labels into account

So, the new space may not be good for classification problem

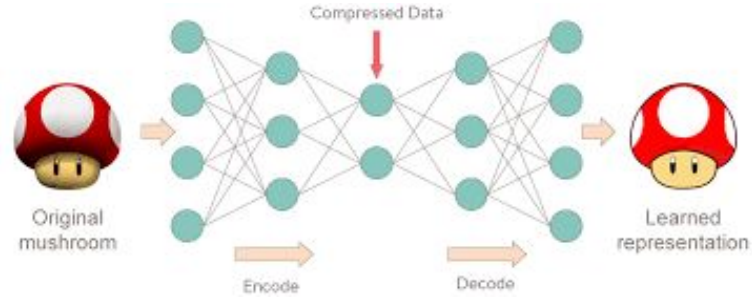
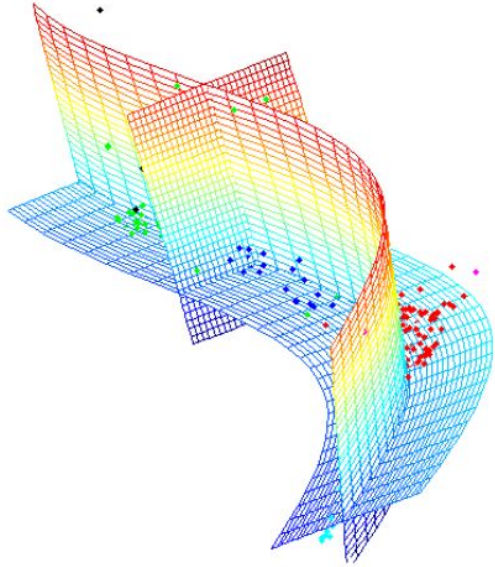


* 2nd P.C. is better than 1st P.C.

Non-linear Transformation can give
better results



Autoencoder networks and Kernel PCA can find these kind of non-linear mappings..



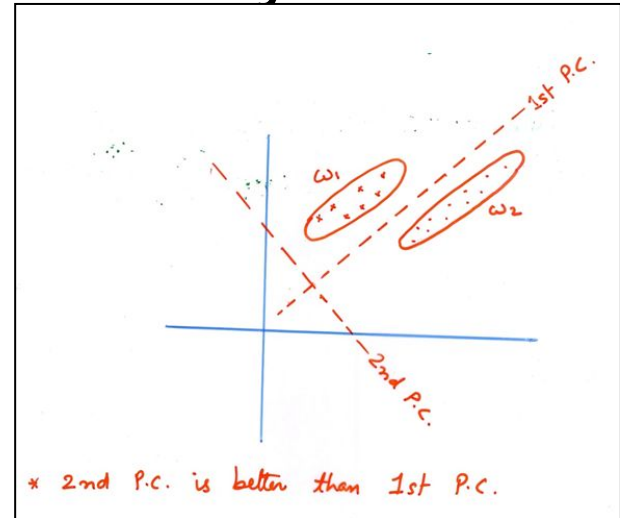
Autoencoder networks and Kernel PCA can find these kind of non-linear mappings..

PCA is unsupervised

- With a labeled (training) data set, for the classification problem, principal components based dimensionality reduction may be bad.

PCA is unsupervised

- With a labeled (training) data set, for the classification problem, principal components based dimensionality reduction may be bad.



Problems and limitations

- What if very large dimensional data?
 - e.g., Images ($d \geq 10^4$)
- Problem:
 - Covariance matrix Σ is size (d^2)
 - $d = 10^4$, then $|\Sigma| = 10^8$
- Singular Value Decomposition (SVD)!
 - efficient algorithms available (Matlab)
 - some implementations find just top N eigenvectors

Singular Value Decomposition

For an $n \times d$ matrix $\mathbf{X}$ of **rank** r there exists a factorization (Singular Value Decomposition = **SVD**) as follows:

$$\mathbf{X} = \mathbf{U} \mathbf{\Lambda} \mathbf{V}^T$$

Diagram illustrating the dimensions of the matrices in the SVD equation:

- $\mathbf{U}$ is $n \times n$
- $\mathbf{\Lambda}$ is $n \times d$
- $\mathbf{V}$ is $d \times d$

The columns of $\mathbf{U}$ are orthogonal eigenvectors of $\mathbf{X}\mathbf{X}^T$.

The columns of $\mathbf{V}$ are orthogonal eigenvectors of $\mathbf{X}^T\mathbf{X}$.

Eigenvalues $\lambda_1 \dots \lambda_r$ of $\mathbf{X}\mathbf{X}^T$ are the eigenvalues of $\mathbf{X}^T\mathbf{X}$.

$$\sigma_i = \sqrt{\lambda_i}$$

$$\mathbf{\Lambda} = \text{diag}(\sigma_1 \dots \sigma_r) \leftarrow \text{Singular values.}$$

SVD example

$$\text{Let } X = \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Thus $n=3$, $d=2$. Its SVD is

$$\begin{bmatrix} 2/\sqrt{6} & 0 & 1/\sqrt{3} \\ -1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & -1/\sqrt{3} \end{bmatrix} \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

Typically, the singular values arranged in decreasing order.

Singular Value Decomposition

- Illustration of SVD dimensions and sparseness

The top diagram illustrates the SVD of a matrix A (5 rows, 3 columns) into three matrices: U (5 rows, 4 columns), Λ (4 rows, 3 columns), and V^T (3 rows, 3 columns). The matrix A is shown as a 5x3 grid of stars. The matrix U is shown as a 5x4 grid of stars, with the last column highlighted in yellow. The matrix Λ is shown as a 4x3 grid of stars, with the last row highlighted in yellow. The matrix V^T is shown as a 3x3 grid of stars. The bottom diagram illustrates the SVD of a matrix A (5 rows, 6 columns) into three matrices: U (5 rows, 3 columns), Λ (3 rows, 4 columns), and V^T (4 rows, 6 columns). The matrix A is shown as a 5x6 grid of stars. The matrix U is shown as a 5x3 grid of stars. The matrix Λ is shown as a 3x4 grid of stars, with the last column highlighted in yellow. The matrix V^T is shown as a 4x6 grid of stars, with the last row highlighted in yellow.

Note: Data Matrix is **A**

Low-rank Approximation

- Solution via SVD

$$A_k = U \operatorname{diag}(\sigma_1, \dots, \sigma_k, \underbrace{0, \dots, 0}_{\substack{\text{set smallest } r-k \\ \text{singular values to zero}}}) V^T$$

$$\underbrace{\begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{bmatrix}}_{A_k} = \underbrace{\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \end{bmatrix}}_U \underbrace{\begin{bmatrix} \bullet & & \\ & \bullet & \\ & & \bullet \end{bmatrix}}_{\Sigma} \underbrace{\begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{bmatrix}}_{V^T}$$

$$A_k = \sum_{i=1}^k \sigma_i u_i v_i^T \longleftarrow \text{column notation: sum of rank 1 matrices}$$

Singular Value Decomposition

- Problem:
 - #1: Find concepts in text
 - #2: Reduce dimensionality

term	data	information	retrieval	brain	lung
document					
CS-TR1	1	1	1	0	0
CS-TR2	2	2	2	0	0
CS-TR3	1	1	1	0	0
CS-TR4	5	5	5	0	0
MED-TR1	0	0	0	2	2
MED-TR2	0	0	0	3	3
MED-TR3	0	0	0	1	1

SVD decomposition

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 2 & 2 & 2 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} | & | \\ \mathbf{u}_1 & \mathbf{u}_2 \\ | & | \end{bmatrix} \times \begin{bmatrix} \lambda_1 & \text{ } \\ \text{ } & \lambda_2 \end{bmatrix} \times \begin{bmatrix} \text{---} & \mathbf{v}_1 & \text{---} \\ \text{---} & \mathbf{v}_2 & \text{---} \end{bmatrix}$$

SVD - Interpretation

‘documents’, ‘terms’ and ‘concepts’:

- **U**: document-to-concept similarity matrix
- **V**: term-to-concept similarity matrix
- **Λ** : its diagonal elements: ‘strength’ of each concept

Projection:

- best axis to project on: (‘best’ = min sum of squares of projection errors)

SVD - Example

- $\mathbf{A} = \mathbf{U} \mathbf{\Lambda} \mathbf{V}^T$ - example:

$$\begin{array}{c}
 \uparrow \\
 \text{CS} \\
 \downarrow \\
 \uparrow \\
 \text{MD} \\
 \downarrow
 \end{array}
 \begin{array}{ccccc}
 & \text{data} & \text{retrieval} & \text{inf.} & \text{brain} & \text{lung} \\
 & & \downarrow & & & \\
 \begin{bmatrix}
 1 & 1 & 1 & 0 & 0 \\
 2 & 2 & 2 & 0 & 0 \\
 1 & 1 & 1 & 0 & 0 \\
 5 & 5 & 5 & 0 & 0 \\
 0 & 0 & 0 & 2 & 2 \\
 0 & 0 & 0 & 3 & 3 \\
 0 & 0 & 0 & 1 & 1
 \end{bmatrix}
 & = &
 \begin{bmatrix}
 0.18 & 0 \\
 0.36 & 0 \\
 0.18 & 0 \\
 0.90 & 0 \\
 0 & 0.53 \\
 0 & 0.80 \\
 0 & 0.27
 \end{bmatrix}
 & \times &
 \begin{bmatrix}
 9.64 & 0 \\
 0 & 5.29
 \end{bmatrix}
 & \times &
 \begin{bmatrix}
 0.58 & 0.58 & 0.58 & 0 & 0 \\
 0 & 0 & 0 & 0.71 & 0.71
 \end{bmatrix}
 \end{array}$$

SVD - Example

- $A = U \Lambda V^T$ - example:

doc-to-concept
similarity matrix

CS-concept MD-concept

↑ CS
↓ MD

$$\begin{bmatrix}
 \text{data} & \text{inf.} & \text{retrieval} & \text{brain} & \text{lung} \\
 1 & 1 & 1 & 0 & 0 \\
 2 & 2 & 2 & 0 & 0 \\
 1 & 1 & 1 & 0 & 0 \\
 5 & 5 & 5 & 0 & 0 \\
 0 & 0 & 0 & 2 & 2 \\
 0 & 0 & 0 & 3 & 3 \\
 0 & 0 & 0 & 1 & 1
 \end{bmatrix}
 =
 \begin{bmatrix}
 0.18 & 0 \\
 0.36 & 0 \\
 0.18 & 0 \\
 0.90 & 0 \\
 0 & 0.53 \\
 0 & 0.80 \\
 0 & 0.27
 \end{bmatrix}
 \times
 \begin{bmatrix}
 9.64 & 0 \\
 0 & 5.29
 \end{bmatrix}
 \times
 \begin{bmatrix}
 0.58 & 0.58 & 0.58 & 0 & 0 \\
 0 & 0 & 0 & 0.71 & 0.71
 \end{bmatrix}$$

SVD - Example

- $A = U \Lambda V^T$ - example:

‘strength’ of CS-concept

↑
CS
↓

↑
MD
↓

	data	inf.	retrieval ↓ brain	lung
1	1	1	0	0
2	2	2	0	0
3	1	1	0	0
4	5	5	0	0
5	0	0	0	2
6	0	0	0	2
7	0	0	3	3
8	0	0	0	1

=

0.18	0
0.36	0
0.18	0
0.90	0
0	0.53
0	0.80
0	0.27

X

9.64	0
0	5.29

X

0.58	0.58	0.58	0	0
0	0	0	0.71	0.71

SVD - Example

- $A = U \Lambda V^T$ - example:

term-to-concept
similarity matrix

↑ CS
↓
↑ MD
↓

$$\begin{bmatrix}
 \text{data} & \text{inf.} & \text{retrieval} & \text{brain} & \text{lung} \\
 1 & 1 & 1 & 0 & 0 \\
 2 & 2 & 2 & 0 & 0 \\
 1 & 1 & 1 & 0 & 0 \\
 5 & 5 & 5 & 0 & 0 \\
 0 & 0 & 0 & 2 & 2 \\
 0 & 0 & 0 & 3 & 3 \\
 0 & 0 & 0 & 1 & 1
 \end{bmatrix}
 =
 \begin{bmatrix}
 0.18 & 0 \\
 0.36 & 0 \\
 0.18 & 0 \\
 0.90 & 0 \\
 0 & 0.53 \\
 0 & 0.80 \\
 0 & 0.27
 \end{bmatrix}
 \begin{bmatrix}
 9.64 & 0 \\
 0 & 5.29
 \end{bmatrix}
 \begin{bmatrix}
 0.58 & 0.58 & 0.58 & 0 & 0 \\
 0 & 0 & 0 & 0.71 & 0.71
 \end{bmatrix}$$

CS-concept

X

X

* (0.58 is circled in the final matrix)

SVD – Dimensionality reduction

- Q: how exactly is dim. reduction done?
- A: set the smallest singular values to zero:

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 2 & 2 & 2 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0.18 & 0 \\ 0.36 & 0 \\ 0.18 & 0 \\ 0.90 & 0 \\ 0 & 0.53 \\ 0 & 0.80 \\ 0 & 0.27 \end{bmatrix} \times \begin{bmatrix} 9.64 & 0 \\ 0 & 5.29 \end{bmatrix} \times \begin{bmatrix} 0.58 & 0.58 & 0.58 & 0 & 0 \\ 0 & 0 & 0 & 0.71 & 0.71 \end{bmatrix}$$

The diagram illustrates the SVD decomposition of a 7x5 matrix. The first matrix is the original data matrix. The second matrix is the orthogonal matrix U (7x7), with the last two columns (0.53, 0.80, 0.27) crossed out with a blue X, indicating they are the smallest singular values being set to zero. The third matrix is the diagonal matrix of singular values (2x2), with the second value (5.29) crossed out with a green X. The fourth matrix is the orthogonal matrix V (2x5), with the last two columns (0.71, 0.71) crossed out with a blue X, indicating they are the smallest singular values being set to zero.

SVD - Dimensionality reduction

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 2 & 2 & 2 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 0.18 \\ 0.36 \\ 0.18 \\ 0.90 \\ 0 \\ 0 \\ 0 \end{bmatrix} \times \begin{bmatrix} 9.64 \end{bmatrix} \times \begin{bmatrix} 0.58 & 0.58 & 0.58 & 0 & 0 \end{bmatrix}$$

SVD - Dimensionality reduction

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 2 & 2 & 2 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 2 & 2 & 2 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$